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Electron-Ion Collider
Record of Decision

TITLE	Decision and Rationale for Selecting the Normal-Conducting 750 MeV S-Band Linac as the RCS Electron Preinjector
PREPARER	Kevin Smith
NUMBER	EIC-ROD-034
DATE	April 07, 2025
AFFECTED WBS/PROJECT AREA	WBS 6.03 Electron Injector
STATEMENT OF DECISION (Summary, 1-2 sentences):	
A decision has been made to select a normal-conducting RF (NCRF) 750 MeV S-Band linac as the RCS electron preinjector linac.	

1. Description/Purpose:

This Record of Decision formalizes the down select decision for the preferred approach of the RCS preinjection linac. EIC-ROD-032 "Decision and Rational for Removing the EIC RCS from the RHIC and EIC Tunnel" [5.1] documented the decision to remove the RCS from the RHIC/EIC tunnel. That decision necessitated a new analysis of RCS electron preinjector options.

Two alternatives were considered:

a. A 750-MeV pulsed super-conducting RF (SRF) linac (L-Band, 1.3 GHz)

b. A 750 MeV pulsed normal-conducting RF (NCRF) linac (S-Band, 2.856 GHz or 2.999 GHz).

The linac energy of 750 MeV was chosen to ensure that the RCS bending dipole fields would stay above ~250 G, which is a comfortable value for magnet design and fabrication. This injection energy is also deliberately chosen to avoid depolarizing resonances, which occur when the spin tune, $G\gamma$, approaches an integer value ($G\gamma \approx 1.7$ for 750 MeV).

2. Note on the Beam Accumulator Ring (BAR):

It was realized during the linac down select analysis that regardless of the choice of linac technology, a Beam Accumulator Ring (BAR) would provide the lowest risk approach to achieving the required 28nC maximum charge per bunch required by the RCS. The decision to use a Beam Accumulator Ring significantly reduced the minimum bunch charge requirement for the linac preinjector. Apart from the requirement for delivering highly polarized electrons, the linac preinjector becomes fairly conventional, similar to that used in many ring based light sources. This is a demonstrated approach used at other facilities. Previous design approaches to implement bunch charge accumulation in the RCS ring carried high technical uncertainty and risk. These approaches were abandoned in favor of a Beam Accumulator Ring located between the preinjector linac and the RCS. The BAR implementation is also planned to include a bypass line

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to provide capability for direct injection of the RCS from the linac preinjector. The BAR decision and implementation is not the subject of this RoD.

3. Advantages and Disadvantages of Each Alternative

a. **L-Band SRF 1.3 GHz Linac:** The primary advantage of an L-Band SRF linac preinjector is its ability to deliver very high bunch charge while preserving required energy spread (due to significantly lower RF curvature and wake induced energy spread compared to S-Band). The BAR obviates the need for such high charge capability, though the ability of such a linac to meet RCS direct injection requirements for operations requiring from 7nC to 11nC per bunch has obvious appeal. The primary disadvantages are the cost and complexity of the associated design, procurement, construction, installation, commissioning, and operation. The cost differential documented in Section 4 is too large to justify the high charge capability.

b. **S-Band NCRF 2.856 GHz Linac:** The primary advantages of an NCRF S-Band linac include the relative simplicity compared to the SRF approach, wide availability of commercial component vendors, and significantly lower capital and operating cost. Cost, schedule and technical risks are all lower than an SRF implementation. System interfaces are cleaner and simpler. Commissiong is simpler and importantly there is no dependence on EIC cryogenic systems (which depend on the RHIC/EIC main cryogenic plant and HSR cryogenic magnet strings being operational). The primary disadvantage is a relatively low limit for maximum single bunch charge of 1nC-2nC, necessary in order to maintain required energy spread and transverse emittance. This significantly limits the RCS direct injection capability and versatility, but such a direct injection mode is not the primary operational configuration for EIC.

4. Cost Comparison:

A careful comparison of S-Band NCRF and L-Band SRF options was carried out to ensure the costs for each solution were understood with reasonable accuracy (-20%, +40%). Details of the comparison are found in the LINAC Full Cost Comparison [5.2]. At the time of this down select, the L-Band SRF system cost range exceeded the S-Band range by a minimum of $44.6M and a maximum of $53.4M.

5. References

5.1 Decision and Rationale for Removing the EIC RCS from the RHIC and EIC Tunnel.pdf

5.2 LINAC Full Cost Comparison_SRF vs NC.xlsx

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APPROVALS:

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